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A MAJOR PROJECT PROPOSAL ON

IoT-Enabled Smart Agriculture Monitoring and Decision Support System

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A MAJOR PROJECT PROPOSAL SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF BACHELOR IN COMPUTER ENGINEERING

Submitted To:
School of Engineering

November 28, 2025

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to Er. Toran Prasad Bhatt, Head of the Department, School of Engineering, Far Western University, for providing us with the platform to present our ideas and the continuous encouragement throughout this proposal. His guidance and support created an environment where we could confidently work, explore, and refine our concept.

We also extend our heartfelt thanks to Birodh Rijal Sir, whose initial orientation and abstract idea helped us visualize the foundation of this project. His suggestions gave us the direction we needed to move forward.

Our sincere appreciation goes to Rishi K. Marseni Sir, Dr. Prakash Datta Sir, Shiv Raj Pant Sir, Er. Kamal Lekhak Sir, Er. Birendra Singh Dhami Sir, Er. Guru Prasad Lekhak Sir, and Er. Rohit Bisht Sir for their valuable insights, thoughtful comments, and continuous encouragement during the formulation of our idea and the understanding of different technical and practical aspects of the system. Their collective experience truly shaped our project into a more meaningful and refined form.

Lastly, we would like to thank all our classmates for their cooperation, support, and motivation throughout this work. Their constant help and team spirit made this journey smoother and more enjoyable.

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1. Introduction

1.1 Background

Agriculture contributes to about one-third of gross domestic product (GDP) in Nepal reaching nearly Rs 57.48 billion in the current fiscal year 2082/83 BS and provides employment to 74% of the economically active population.

Despite this, the reality on the ground paints a starkly different picture, one where farmers still struggle with old challenges such as erratic weather, pests, (poor infrastructure). Climate change has brought unpredictable hailstorms, floods, and unseasonal rains that destroy crops and reduce yields. Farmers often rely on intuition and other irrelevant factors, such as the desire for quick profits or ignorance of market demands, when deciding which crops to cultivate. This can lead to poor choices and financial strain on farmers, leading to discouragement from the profession.

Agriculture remains the backbone of many economies, yet it faces unprecedented challenges: unpredictable weather, water shortage, soil degradation, pest attacks, and lack of timely expert advice. According to the United Nations, food production must increase by 60–70% by 2050 to feed 10 billion people. Traditional methods rely heavily on experience and guesswork, leading to over-irrigation (wasting up to 50% water), improper fertilizer use, and delayed response to crop stress.

The convergence of IoT, cloud computing, and AI offers a transformative solution. Low-cost sensors can continuously monitor soil and weather parameters, cloud platforms enable storage and real-time analytics, and machine learning models can predict optimal actions with high accuracy. This project implements exactly such an integrated system.

1.2 Problem Statement

Problems in the current agriculture system are:

- Dependency on weather and unverified information.
- Intuition-based decisions.
- Lack of real time field data.

- Illiteracy about soil characteristics.
- Reduced crop productivity.

These issues make traditional farming less productive, less profitable, and less sustainable in Nepal. So, our proposed system will mitigate these problems and help farmers to make data driven decisions rather than intuition based imprecise decisions.

1.3 Objectives

The main objectives of this project are:

- Collect real-time data (temperature, humidity, soil moisture, pH) using sensors and transmit to cloud via ESP32.
- Implement ML models to analyze data and help make data-driven decisions.
- Provide recommendations for crops, fertilizers and irrigation.
- To make farming smart and sustainable.

2. Research Methodology

2.1 Literature Review

The integration of Internet of Things (IoT), cloud computing, and artificial intelligence (AI) to transform traditional agriculture into a fully automated, data-driven system has gained significant momentum in recent years.

Alreshidi (2019) highlighted the fragmentation issues in current agricultural processes and proposed a comprehensive Smart Sustainable Agriculture (SSA) platform underpinned by IoT and AI that addresses sensor data collection, cloud storage, big data analytics, interoperability, and real-time decision support – a vision that directly aligns with the architecture of the proposed system [1].

Quy et al. (2022) presented a detailed survey of IoT-enabled smart agriculture ecosystems, emphasizing a layered architecture comprising perception (sensors/WSN), network, cloud processing, big data analytics, and application layers. They stressed the necessity of real-time monitoring of soil moisture, temperature, humidity, pH, nutrient levels, and weather data, and pest detection, followed by AI-driven predictive analytics and farmer-facing interfaces – essentially the exact blueprint this project follows [2].

A 2024 preprint systematically reviewed the fusion of Wireless Sensor Networks, IoT, Cloud Computing, and AI/ML for smart farming, proposing a modular architecture that includes (1) WSN for data acquisition, (2) edge/cloud processing, (3) AI models for crop recommendation, irrigation scheduling, yield prediction, and crop rotation planning, and (4) user-friendly web/mobile interfaces for real-time visualization and control. The authors explicitly state that such integrated systems lead to higher crop quality, optimal resource use, and weather-responsive strategies, validating the core objectives of the present work [3].

Ibrahim et al. (2025) demonstrated the practical feasibility of combining IoT with deep learning (ResNet50 achieving 99.8% accuracy) for automated plant disease detection and treatment using low-cost hardware mounted on existing irrigation pivots, proving that high-accuracy AI models can run locally or via cloud with minimal latency and provide immediate actionable outputs (e.g., targeted spraying), which strengthens the case for including pest/disease prediction modules in the proposed platform [4].

These studies collectively confirm that the convergence of IoT sensors, cloud-based storage and analytics, AI/ML for predictive insights (crop suitability, irrigation/fertilization scheduling, yield forecasting), and intuitive web/mobile dashboards represents the current state-of-the-art and the most promising pathway to sustainable, profitable, and climate-resilient agriculture – precisely the foundation upon which this project is built.

2.2 Requirement Specification

2.2.1 Functional Requirements

- Real-time collection of soil and environmental parameters, including pH, soil moisture, temperature, and humidity, using IoT sensors.
- Transmission of sensor data to a FastAPI backend via WiFi (ESP32) or LoRa for long-range communication.
- Storage of raw sensor readings and processed predictions in a database.
- Execution of ML models to provide crop recommendations and fertilizer suggestions based on soil and weather data.

- Alerts and notifications for soil anomalies, crop suitability, and irrigation/fertilizer needs.
- Visualization of live sensor readings, predictions, and historical data through a frontend dashboard accessible to farmers.

2.2.2 Non-Functional Requirements

- Reliability: Continuous and accurate sensor monitoring.
- Scalability: Support multiple sensors and farms without performance loss.
- Real-time Performance: Quick sensor processing and ML predictions.
- Usability: Clean, user-friendly dashboard for farmers.
- Maintainability: Easy updates for ML models, sensors, and features.
- Availability: High uptime with minimal downtime.

2.2.3 System Requirements

The approximate hardware requirement is described in the table below:

Table 1: Hardware requirements description

Component	Purpose	Justification	Possible Replacements / Complements
ESP32 Microcontroller	Collects sensor data and transmits it to backend	Low cost, built-in WiFi/Bluetooth, easy to program	Arduino Uno/Nano (for simpler setups), Raspberry Pi Pico W (more computational power)
pH Sensor	Measures soil acidity for crop recommendation	pH affects which crops can grow	Alternative: Atlas Scientific pH probe (more precise)
Soil Moisture Sensor	Monitors water content for irrigation recommendations	Moisture impacts crop growth	Capacitive sensors preferred over resistive sensors for accuracy
BME 280 (Temperature & Humidity)	Measures environmental conditions	Temperature & humidity affect crop suitability	DHT11 or DHT22 (cheaper, less precise)

NPK Sensor	Measures soil nutrient levels (Nitrogen, Phosphorus, Potassium)	Necessary for fertilizer recommendations	DIY ion-selective electrodes (more complex), commercial 3-in-1 NPK sensors
GPS Module (Optional)	Captures farm location for region-specific predictions	Crop suitability depends on geographic location	None, or use manual input of farm location
LoRa Module (Optional)	Long-range communication for remote farms	Useful if farm is far from WiFi	WiFi mesh network, NB-IoT modules
Power Supply / Solar Panel	Ensures continuous operation in field	IoT devices need stable power	Rechargeable batteries, UPS for indoor setups
Enclosure / Waterproof Box	Protects electronics in outdoor environment	Prevents damage from weather	Plastic or metal enclosures, weatherproof coating

Software & Technologies

- Firmware : Arduino IDE / MicroPython
- Cloud : ThingSpeak (free tier) or AWS IoT Core + DynamoDB
- Backend : Python FastAPI or Node.js
- Database : MongoDB / Firebase Realtime DB
- AI/ML : Python (scikit-learn, TensorFlow/Keras)
- Frontend : React.js + Shadcn
- Communication: MQTT protocol

2.3. Feasibility Study

2.3.1 Technical Feasibility

From a technical point of view, this project is completely feasible because it uses technologies and tools that are already familiar to engineering students and are easy to learn within the project timeline. The sensors we are using such as pH, NPK, moisture, temperature, and humidity are all fully compatible with common microcontrollers like ESP32, so the technical setup does not require any advanced industrial hardware. The

programming required involves basic sensor interfacing, simple IoT data transmission, a backend to receive and process the data, and machine learning models that run on Python, which we already use in our academics. Since all the software we need FastAPI, Python, PostgreSQL or Firebase, and dashboard frameworks is free and open source, there is no technical barrier to implementation. The entire architecture (IoT → backend → database → dashboard → predictions) is based on standard practices that we can comfortably implement with available resources. For these reasons, the project is technically feasible and realistic for a final-year major project.

2.3.2 Operational Feasibility

Operationally, the system is simple to run and does not require any specialized training. Once the sensors are installed in the field, the ESP32 automatically sends the readings to the backend, where the machine learning model processes them and the dashboard displays the recommendations. Even someone with basic computer knowledge can operate the dashboard, understand the readings, and follow the recommendations for crop selection, fertilizer usage, and irrigation planning. Maintenance is also easy because individual sensors can be replaced or recalibrated without affecting the whole system, and the software components can be updated or improved at any time. The system can run continuously as long as it has a power source and internet connection, which makes it suitable not only for academic demonstration but also for small-scale real-world deployment. Because of this smooth day-to-day usability, the project is operationally feasible and practical for both students and eventual end-users like farmers

2.3.3 Economic Feasibility

This project is economically feasible as most of the required components are affordable and easily available in Nepal, especially through platforms like Daraz. The School of Engineering also plays an important role here, because the availability of labs, microcontrollers, computer systems, and internet connection significantly reduces our overall cost. The only slightly expensive components in this project are the NPK sensor and the pH sensor, but these are also the most important ones. These two sensors directly

determine how accurate and scientific our fertilizer and crop recommendations will be. Without them, the system cannot give precise output, but with them we can make highly precise and reliable predictions that truly help farmers. This alone justifies the investment in these sensors. Availability of these devices will allow us to build a complete working prototype that can create impact and can also be used for future research works. Overall, the cost of the project is feasible compared to the usefulness and outcomes it delivers.

Table 2: Cost estimation

Components	**Cost (NPR)**
ESP32	900 – 1,400
Soil Moisture Sensor	300 – 500
BME280 Sensor	800 – 1,200
Soil pH Sensor	3,000 – 4,500
NPK Sensor	8,000 – 9,500
Power Supply	150
Jumper Wires + Misc	300 – 600

Total Estimated Cost: NPR 12,000 - 16,000.

2.4 Model Used

In this project, we will follow an incremental methodology so that every part of the system becomes manageable. First, we will study how each sensor works and then test them individually using an ESP32, because this will help us understand the accuracy of pH, moisture, temperature, and humidity readings before combining them. After verifying stability, we will integrate all sensors together and set up the ESP32 to send data to our backend.

Next, we will develop a FastAPI backend that will collect sensor data and store it in the database. We will start with dummy data so that we can test our API structure without depending on hardware. Once the backend becomes stable, we will connect the ESP32 to send real sensor readings.

For the machine learning part, we will collect and preprocess agricultural datasets and then train simple ML models for crop, fertilizers, and irrigation recommendations. We will choose models like Random Forest and Regression because they are beginner-friendly and give reliable results with limited data. The trained model will then be integrated into the backend so that predictions can be generated automatically.

Finally, we will build a dashboard to display sensor values and recommendations to farmers. The dashboard will fetch processed data from the backend and present it in a clear and user-friendly way. Throughout the development, we will follow a step-by-step testing approach so that each module works properly before moving to the next stage.

3. High Level Design of the System

3.1 System Overview

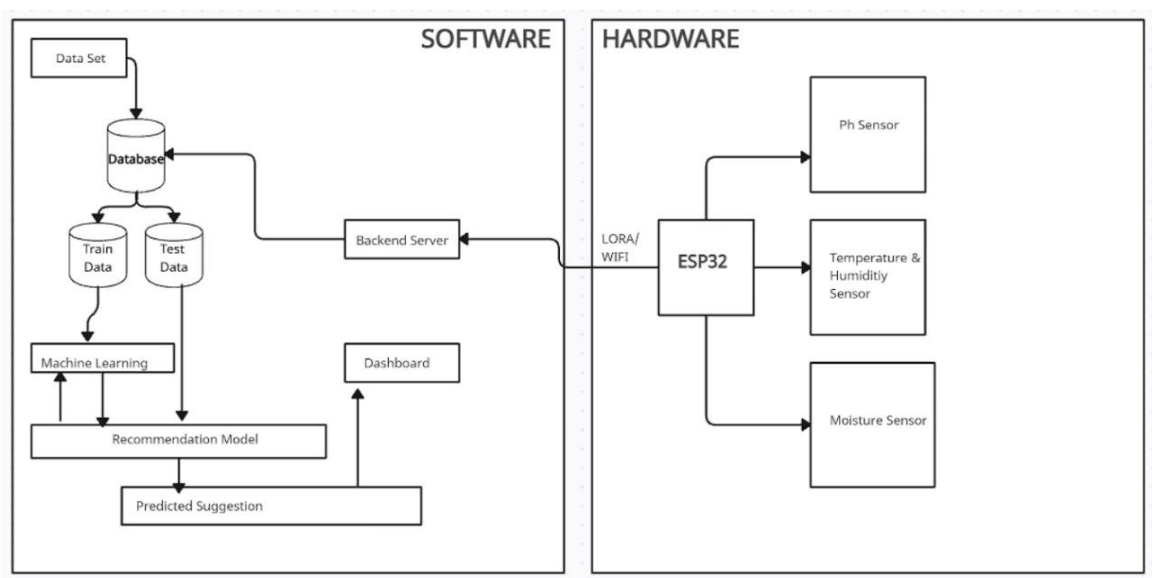

Figure 1: Overall System Block Diagram

The block diagram above represents the possible architecture of our system. It shows how sensors, controllers, data processing models and user interface interact with each other to make a functional system.

3.2 System Flow Diagram

The flowchart of our system is drawn below:

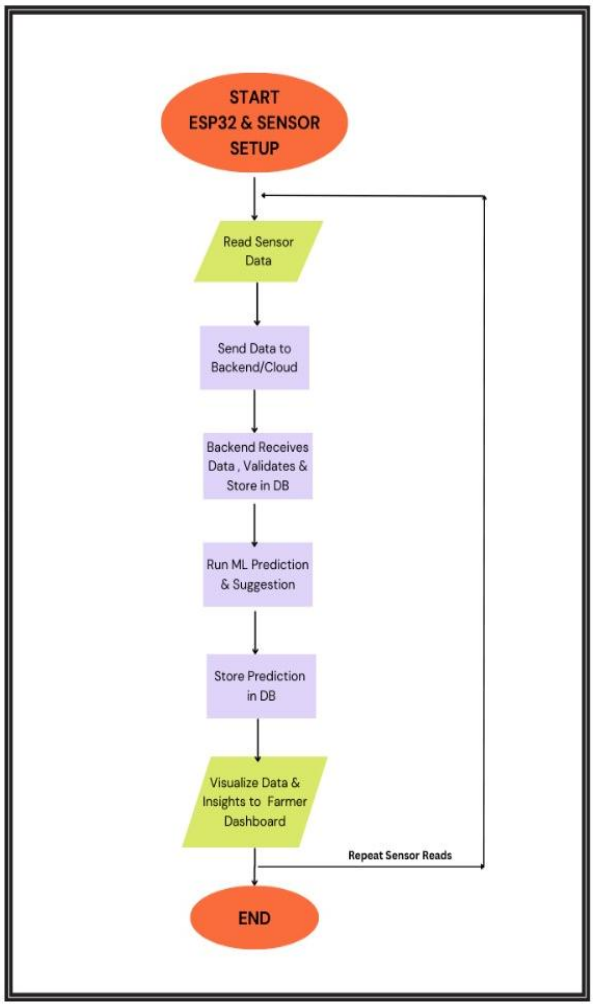

Figure 2: Basic workflow of our system

Sensors take data readings (temperature, humidity, moisture, pH and N, P, K content) from physical surroundings. Then the collected data is sent to backend of the system which then receives and validates the input data and then stores in database. Then the ML model is fed

with the data and predictions will be made on crop, fertilizer and irrigation recommendation. The predictions are then stored in database and used to display insights to farmer's dashboard.

4. Algorithm & Mathematical Model

Random Forest is a machine learning ensemble algorithm that builds many decision trees and combines their outputs to make a final prediction. Random Forest models are highly beneficial for smallholder farms in developing countries due to their ability to handle complex and noisy data, which is common in agricultural contexts. These models excel at capturing nonlinear relationships between variables such as weather patterns, soil characteristics, crop types, and yields, providing accurate predictions and insights for farmers. The ensemble learning approach of Random Forests mitigates overfitting and improves generalization, making them suitable for situations with limited data availability. Following steps are involved in a Random Forest Algorithm:

1. For b=1 to N=B:
    a) Draw a bootstrap sample Z^* of size N from the training data.
    b) Grow a random-forest tree T_b to the bootstrapped data, by recursively repeating the following steps for each terminal node of the tree, until the minimum node size n_{min} is reached.
        i) Select m variables at random from the p variables.
        ii) Pick the best variable/split-point among the m.
        iii) Split the node into two daughter nodes.
2. Output the ensemble of trees $\{T_b\}_1^B$.

To make a prediction at a new point x:

Regression: $\hat{f}_{\mathrm{rf}}^B(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x)$.

Classification: Let $\hat{C}_b(x)$ be the class prediction of the bth random-forest tree. Then $\hat{C}_{\mathrm{rf}}^B(x) = majority\ vote\ \{\hat{C}_b(x)\}_1^B$.

5. Expected Outcomes

A fully functional web-based platform will be developed to assist users in accessing legal

- Live dashboard showing temperature, moisture, pH, NPK ratios.
- Crop recommendation with confidence score.
- Daily irrigation schedule (e.g., “Irrigate 25 mm today at 6 AM”).
- Fertilizer suggestion (e.g., “Apply 30 kg Urea/ha”).
- SMS/app alerts or notification: “Moisture low → Starting pump”.
- Accuracy target: ≥92% on crop recommendation, ≥95% irrigation savings vs traditional.

6. Project Timeline

Table 3: Gantt Chart (Nov 2025 - May 2026)

Task	Nov	Dec	Jan	Feb	Mar	Apr	May
Literature Review & Proposal	■	■					
Hardware Procurement & Setup		■	■				
Sensor Integration & Calibration			■	■			
Cloud Setup & Database			■	■			
AI Model Development			■	■	■	■	
Web + Mobile App Development			■	■	■	■	
Field Testing & Validation						■	■
Documentation & Presentation	■				■		■

7. References

[1] E. Alreshidi, “Smart Sustainable Agriculture (SSA) Solution Underpinned by Internet of Things (IoT) and Artificial Intelligence (AI),” Int. J. Adv. Comput. Sci. Appl., vol. 10, no. 5, pp. 93–102, 2019.

[2] V. K. Quy, N. V. Hau, D. V. Anh, N. M. Quy, N. T. Ban, S. Lanza, G. Randazzo, and A. Muzirafuti, “IoT-Enabled Smart Agriculture: Architecture, Applications, and Challenges,” Appl. Sci., vol. 12, no. 7, p. 3396, 2022, doi: 10.3390/app12073396.

[3] “Optimizing Agricultural Efficiency: A Fusion of IoT, AI, Cloud Computing, and Wireless Sensor Network,” Preprint, SSRN Electron. J., 2024, doi: 10.2139/ssrn.4789232.

[4] A. S. Ibrahim et al., “AI-IoT based smart agriculture pivot for plant diseases detection and treatment,” Sci. Rep., vol. 15, no. 1, p. 16576, Dec. 2025, doi: 10.1038/s41598-025-98454-6.